

42- MAU-001 ENABLED

ALL MOTORIZED DAMPERS SHALL OPEN TO THEIR FULL OPEN POSITION. FINAL DAMPER POSITION SHALL BE ADJUSTABLE.

42- MAU-001 DISABLED

ALL MOTORIZED DAMPERS SHALL BE IN THEIR CLOSED POSITION. FINAL DAMPER POSITION SHALL BE ADJUSTABLE.

TUNNEL ENCLOSURE DAMPER ALARM

IF ANY TUNNEL ENCLOSURE DAMPER FAILS TO OPEN AFTER 60 SEC (ADJ) SEND ALARM TO BMS FRONT END

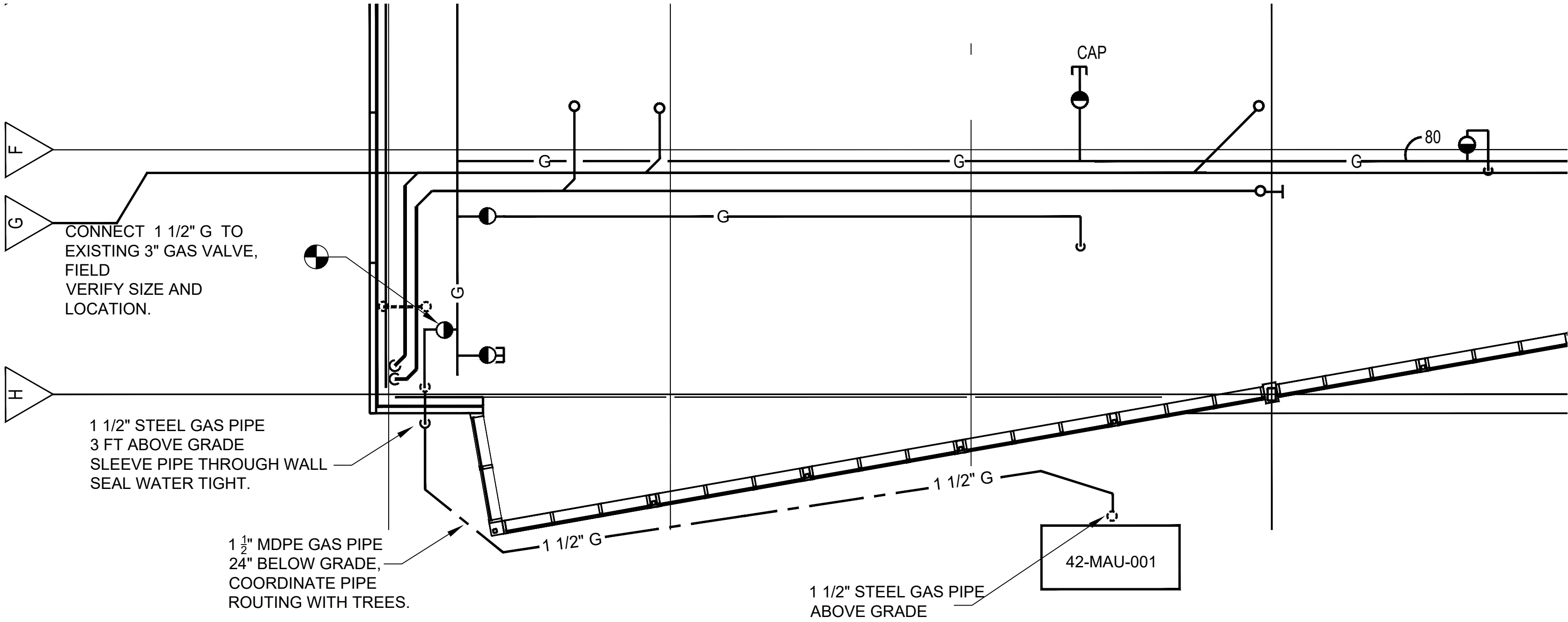
TUNNEL STAIR ENCLOSURE VENTILATION DAMPER SEQUENCE

1. TUNNEL HEATING – ELECTRICAL SCOPE OF WORK
 - a. DISCONNECT THE EXISTING TWO SUPPLY FANS IN THE TUNNEL. EACH FAN IS 7.5 HP, 480V, 3-PHASE.
 - i. ONE OF THE EXISTING CIRCUITS SHALL REMAIN FOR THE CONNECTION TO THE NEW MAKE-UP AIR UNIT.
 - ii. THE OTHER CIRCUIT SHALL BE REMOVED BACK TO IT’S SOURCE WITH THE EXCEPTION OF ITS CONDUIT. CONDUIT SHALL REMAIN FOR FUTURE USE AND BE LABELED AS “SPARE” ON ALL OF ITS ASSOCIATED BOXES. THE CIRCUIT BREAKER SHALL ALSO BE LABELED AS “SPARE”.
 - b. CONNECT NEW MAKE-UP AIR UNIT (5 HP, 480V, 3-PHASE) TO THE EXISTING CIRCUIT THAT REMAINED FROM 1., a., i. RELABEL ITS LOAD IN THE PANELBOARD DIRECTORY. CONFIRM THAT THE MAKE-UP AIR UNIT IS PROVIDED WITH A VFD SINCE THAT VFD WILL SERVE AS ITS DISCONNECTING MEANS. CONFIRM VOLTAGE AND OTHER ELECTRICAL REQUIREMENTS WITH THE MECHANICAL CONTRACTOR. NEW LOAD IS LESS THAN LOAD REMOVED.
 - c. PROVIDE ELECTRICAL CONNECTIONS TO THREE 10kW UNIT HEATERS. EACH UNIT HEATER WILL BE LOCATED AT THE BOTTOM LANDING OF EACH TUNNEL ENTRY POINT.
 - i. EACH UNIT HEATER IS RATED FOR 10kW, 480V, 3-PHASE. CONFIRM VOLTAGE AND OTHER ELECTRICAL REQUIREMENTS WITH THE MECHANICAL CONTRACTOR.
 - ii. PROVIDE NEW 20A3P CIRCUIT BREAKERS IN EXISTING PANELBOARDS “LP1”, “LP2”, AND “LP3”. NEW CIRCUIT BREAKERS SHALL MATCH EXISTING PANELBOARD’S MANUFACTURER AND AIC RATINGS. PANELBOARDS “LP1”, “LP2”, AND “LP3” ARE LOCATED IN THE TUNNEL BELOW EACH TUNNEL ENTRY POINT.
 - iii. CIRCUIT SHALL BE (3#12+ #12G)3/4”C.
 - iv. DISCONNECTING MEANS SHALL BE 30A3P, HEAVY-DUTY SAFETY DISCONNECT.
 - v. COORDINATE EXACT LOCATION OF THE UNIT HEATERS WITH MECHANICAL CONTRACTOR.
 - d. CONTRACTOR SHALL SURVEY EXISTING LOADS ON PANELBOARDS “LP1”, “LP2”, AND “HVDC”, THEN PROVIDE UPDATED AND COMPLETE LOAD SUMMARIES FOR ALL PANELBOARDS, DISTRIBUTION EQUIPMENT, ETC. THAT IS AFFECTED BY THE ADDITIONAL LOADS.
 - e. PROVIDE 30-DAY METERING AS NECESSARY TO CONFIRM PANELBOARD CAPACITIES AND ASSIST IN THE LOAD STUDIES.

ELECTRICAL SPECIFICATIONS AND OTHER MISCELLANEOUS REQUIREMENTS

1. REFER TO PRODUCT DATA OF ALL NEW MECHANICAL EQUIPMENT FOR ELECTRICAL DATA AND ADDITIONAL REQUIREMENTS.
2. PROVIDE NEW ELECTRICAL DISTRIBUTION EQUIPMENT AS NECESSARY FOR A COMPLETE SYSTEM.
3. LABEL ALL NEW EQUIPMENT TO MEET NIST’S SITE STANDARD REQUIREMENTS.
4. PTOVIDE NEW CONDUITS AS NECESSARY. COLOR OF NEW CONDUITS TO MEET NIST’S SITE STANDARD REQUIREMENTS.
5. ALL INSTALLATIONS SHALL MEET NIST’S SITE STANDARD REQUIREMENTS. REFER TO NIST’S SITE STANDARD SPECIFICATIONS ATTACHMENT FOR ADDITIONAL INFORMATION AND REQUIREMENTS.
6. REFER TO ATTACHED DOCUMENTS FOR ADDITIONAL INFORMATION AND REQUIREMENTS.

ELECTRICAL SCOPE OF WORK



42-MAU-001 UNDER GROUND GAS SERVICE

SCALE: NOT TO SCALE

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
BOULDER DESIGN AND CONSTRUCTION DIVISION
325 BROADWAY BOULDER, COLORADO 80305-3328

BUILDING 42 TUNNEL VENTILATION

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REVISION	DESCRIPTION	BY	SUPV.	DATE

DESIGN COOK	DRAFTING COOK	APPROVAL	DATE 3/10/2026	SHEET NUMBER M-3 of
PROJECT NAME ADDITIONAL TUNNEL HEAT				